# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: Artificial Intelligence** **COURSE CODE: CSA1711**

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**COURSE OUTCOME COVERED**

***CO1:*** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*( BL3)*

*Assessment Tool Weightage: 50%*

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**QUESTIONS: 3** **Duration : 1 Hour**
**Total Marks:30**

Annexure A

## Constraint-Based Problem Solving

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***Code of Conduct:***

*I MS Naveen (192371002) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

*Naveen.ms*

Annexure A

# Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

   i. D1 cannot work Night shift ii. D2 must work before D3 (Morning < Afternoon < Night) iii. Only one doctor per shift iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved. c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Problem Understanding | 25% | Clearly defines the problem and identifies all relevant constraints accurately | Identifies most constraints with minor gaps | Partial understanding; misses key constraints | Fails to identify constraints correctly |
| Constraint Analysis | 25% | Analyzes constraints effectively and prioritizes them logically | Provides reasonable analysis with some prioritization | Limited analysis; weak prioritization | No meaningful analysis or prioritization |
| Solution Development | 25% | Develops optimal and feasible solutions satisfying all constraints | Develops workable solutions with minor limitations | Solutions partially satisfy constraints | Solutions do not meet constraints |
| Application of Concepts | 15% | Effectively applies appropriate concepts, methods, or tools | Applies concepts with minor errors | Limited application; requires guidance | Unable to apply relevant concepts |
| Justification | 10% | Provides strong justification for | Provides justification with some clarity | Weak or unclear justification | No justification provided |

| | | | | | |
|---|---|---|---|---|---|
| | | decisions with logical reasoning | | | |

1.A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

i. D1 cannot work Night shift ii. D2 must work before D3 (Morning < Afternoon < Night)

iii. Only one doctor per shift iv. D3 cannot work Morning

v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Backtracking Search Solution

#Step 1: Define Variables and Domains

*Doctors:D1, D2, D3

*Shifts:

* Morning (M)
* Afternoon (A)
* Night (N)

*Constraints:

1. D1 cannot work Night.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor per shift.
4. D3 cannot work Morning.
5. Every doctor is assigned exactly one shift.

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# Step 2: Backtracking Process

#Initial State

* D1 = ?
* D2 = ?
* D3 = ?

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#Assign D1

Try *Morning ✔️

* D1 = Morning

Remaining shifts: Afternoon, Night

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#Assign D2

Try  Afternoon*✔️

* D2 = Afternoon

Remaining shift: Night

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#Assign D3

Only  Night  is available.

Check constraints:

* D3 ≠ Morning ✔️
* D2 works before D3 (Afternoon < Night) ✔️
* One doctor per shift ✔️

All constraints are satisfied.

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# Constraint Checks

| Constraint | Status |
| --- | --- |
| D1 cannot work Night | ✔️ D1 = Morning |
| D2 before D3 | ✔️ Afternoon < Night |
| One doctor per shift | ✔️ |
| D3 cannot work Morning | ✔️ D3 = Night |
| Each doctor assigned one shift | ✔️ |

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#Backtracking Tree

```text
Start
│
├── D1 = Morning ✔
│   │
│   ├── D2 = Afternoon ✔
│   │   │
│   │   └── D3 = Night ✔ (Valid Solution)
```

Since the first assignment satisfies all constraints, **no backtracking is required**.

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# Final Valid Schedule

| Shift | Doctor |
| --- | --- |
| Morning | D1 |
| Afternoon | D2 |
| Night | D3 |

#Final Answer

* Morning → D1
* Afternoon → D2
* Night → D3

This assignment satisfies all the given constraints.

4. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

i. Movement allowed: Up, Down, Left, Right ii. Cost of each move = 1
v. Cannot pass through obstacles
vi. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

#BFS Search Algorithm Solution

Assume the following **5 × 5 grid**.

```text
S . . X .
. X . X .
. X . . .
. . X X .
. . . . G
```

Where:

*S = Start (0,0)

*G= Goal (4,4)

*X = Obstacle

* .= Free cell

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#Step 1: Initial State

*Start: (0,0)

*Goal:(4,4)

*Queue:[(0,0)]

*Visited:{(0,0)}

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#Step 2: BFS Node Expansion

| Step | Expanded Node | Queue after Expansion |
| --- | --- | --- |
| 1 | (0,0) | (1,0), (0,1) |
| 2 | (1,0) | (0,1), (2,0) |
| 3 | (0,1) | (2,0), (0,2) |
| 4 | (2,0) | (0,2), (3,0) |
| 5 | (0,2) | (3,0), (1,2) |
| 6 | (3,0) | (1,2), (4,0), (3,1) |
| 7 | (1,2) | (4,0), (3,1), (2,2) |
| 8 | (4,0) | (3,1), (2,2), (4,1) |
| 9 | (3,1) | (2,2), (4,1) |
| 10 | (2,2) | (4,1), (2,3) |
| 11 | (4,1) | (2,3), (4,2) |
| 12 | (2,3) | (4,2), (2,4) |

| 13 | (4,2) | (2,4), (4,3) |

| 14 | (2,4) | (4,3), (3,4), (1,4) |

| 15 | (4,3) | (3,4), (1,4), (4,4) |

| 16 | (4,4) | Goal Reached |

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# Priority Queue (BFS Queue) Updates

```text
Initial:
[(0,0)]

After expanding (0,0):
[(1,0), (0,1)]

After expanding (1,0):
[(0,1), (2,0)]

After expanding (0,1):
[(2,0), (0,2)]
```

...

Final:

[(4,4)] → Goal Found

```

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#Manhattan Distance Heuristic

Formula:

*$h(n) = |x_1 - x_2| + |y_1 - y_2|$

Example:

For Start (0,0) and Goal (4,4):

$h = |4-0| + |4-0| = 8$

> **Note:** BFS does **not** use the Manhattan heuristic for node selection. The Manhattan distance heuristic is used by informed search algorithms such as **A***.

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#Optimal Path
```

```text
(0,0)
→ (1,0)
→ (2,0)
→ (3,0)
→ (4,0)
→ (4,1)
→ (4,2)
→ (4,3)
→ (4,4)
```

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#Path Cost

* Number of moves = **8**
* Cost per move = **1**

**Total Cost = 8**

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#Final Answer

*Algorithm: Breadth-First Search (BFS)

*Optimal Path:S → (1,0) → (2,0) → (3,0) → (4,0) → (4,1) → (4,2) → (4,3) → G

*Total Cost:8

*Heuristic:Manhattan Distance = (|x_1-x_2| + |y_1-y_2|) (used in A*; not used by BFS).